POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Artificial Intelligence (New) | Pass Marks : 45
Time : 3 hrs

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) If computer passes Turing test, will it act like human? justify with reasons. What are impacts of AI in society? 7

   b) What is PEAS? Explain PEAS representation for an autonomous taxi agent. 8

2. a) Why is searching important in problem solving? What are the drawbacks of Greedy Best First Search and how A* search technique is used to solve it. Explain with example. 7

OR

Why is Hill Climbing called as Local greedy search? Explain the problems in Hill Climbing with appropriate diagram.

   b) what are the constraint satisfaction problems? Solve the following crypt arithmetic problem **BASE+BALL=GAMES** 8

3. a) Explain any four knowledge representation techniques used in AI. 8

   b) Consider the following axioms: 7

   i. Ravi likes all kind of food.
   ii. Apples and chicken are food
   iii. Anything anyone eats and is not killed is food
   iv. Ajay eats peanuts and is still alive
   v. Rita eats everything that Ajay eats.

   Prove by resolution that Ravi likes peanuts using resolution.

4. a) Explain Bayesian Networks with example. 8

   b) What do you understand by Learning by analogy? Explain in detail. 7

5. a) How do you differentiate K means clustering and K nearest neighbor algorithm? Using K-means Clustering, cluster the following data into clusters. Where k= 3 for data: 8

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	A	B
1	2	5
2	3	6
3	5	4
4	7	7
5	3	5
6	5	3

b) Differentiate between Supervised Learning and Unsupervised Learning with example. 7

6. a) What do you mean by membership function in fuzzy logic? Differentiate two types of membership functions: Triangular and Trapezoidal method. 8

OR

Difference between Fuzzification and Defuzzification. Explain in detail.

b) Explain the architecture of expert systems in detail. 7

7. Write short notes on: (Any two) 2×5

a) Alpha-beta pruning

b) Semantic Net and Frames

c) Forward and Backward Chaining

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